

Bitwise differential trail through **TWINE** found by SAT

CiVerLy

July 24, 2026

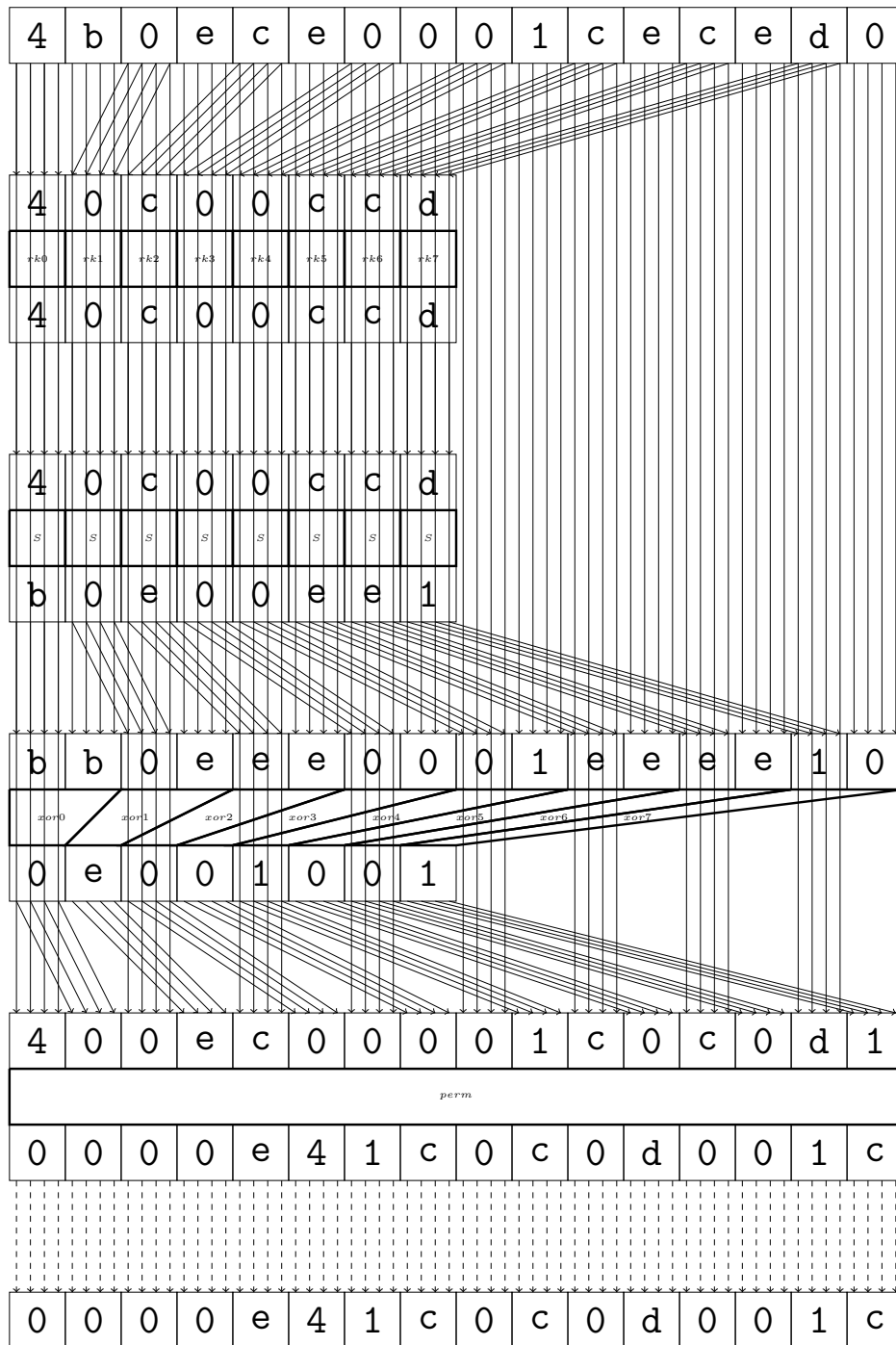
1 TWINE

Maximal differential probability: 2^{-38}

4	b	0	e	c	e	0	0	0	1	c	e	c	e	d	0
4	b	0	e	c	e	0	0	0	1	c	e	c	e	d	0
round															
0	0	0	0	e	4	1	c	0	c	0	d	0	0	1	c
0	0	0	0	e	4	1	c	0	c	0	d	0	0	1	c
round															
0	0	d	1	0	0	c	e	0	0	0	1	0	0	0	0
0	0	d	1	0	0	c	e	0	0	0	1	0	0	0	0
round															
0	d	1	c	0	0	0	0	0	0	0	0	0	0	0	0
0	d	1	c	0	0	0	0	0	0	0	0	0	0	0	0
round															
d	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
d	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
round															
0	0	0	0	0	d	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	d	0	0	0	0	0	0	0	0	0	0
round															
0	0	0	0	0	0	0	0	0	0	0	0	d	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	d	0	0	0
round															
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	d
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	d
round															
0	0	c	0	0	0	0	0	0	1	0	0	0	0	d	0
0	0	c	0	0	0	0	0	0	1	0	0	0	0	d	0
round															
0	c	0	0	e	0	1	0	0	0	0	d	0	0	1	0
0	c	0	0	e	0	1	0	0	0	0	d	0	0	1	0
final															
0	c	0	0	e	4	1	c	0	0	0	d	0	0	1	c
0	c	0	0	e	4	1	c	0	0	0	d	0	0	1	c

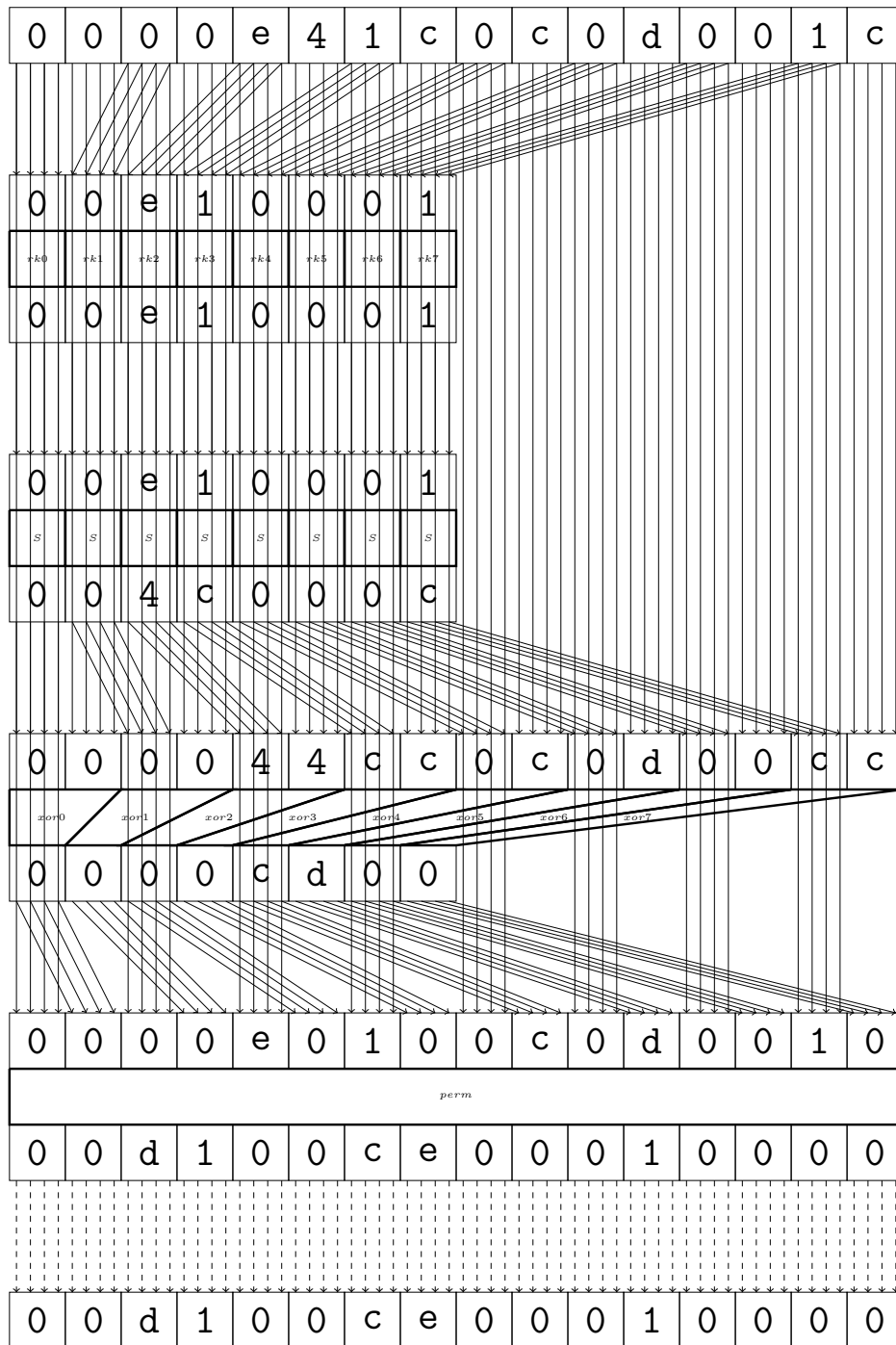
2 round

Maximal differential probability: $2^{-10.0}$



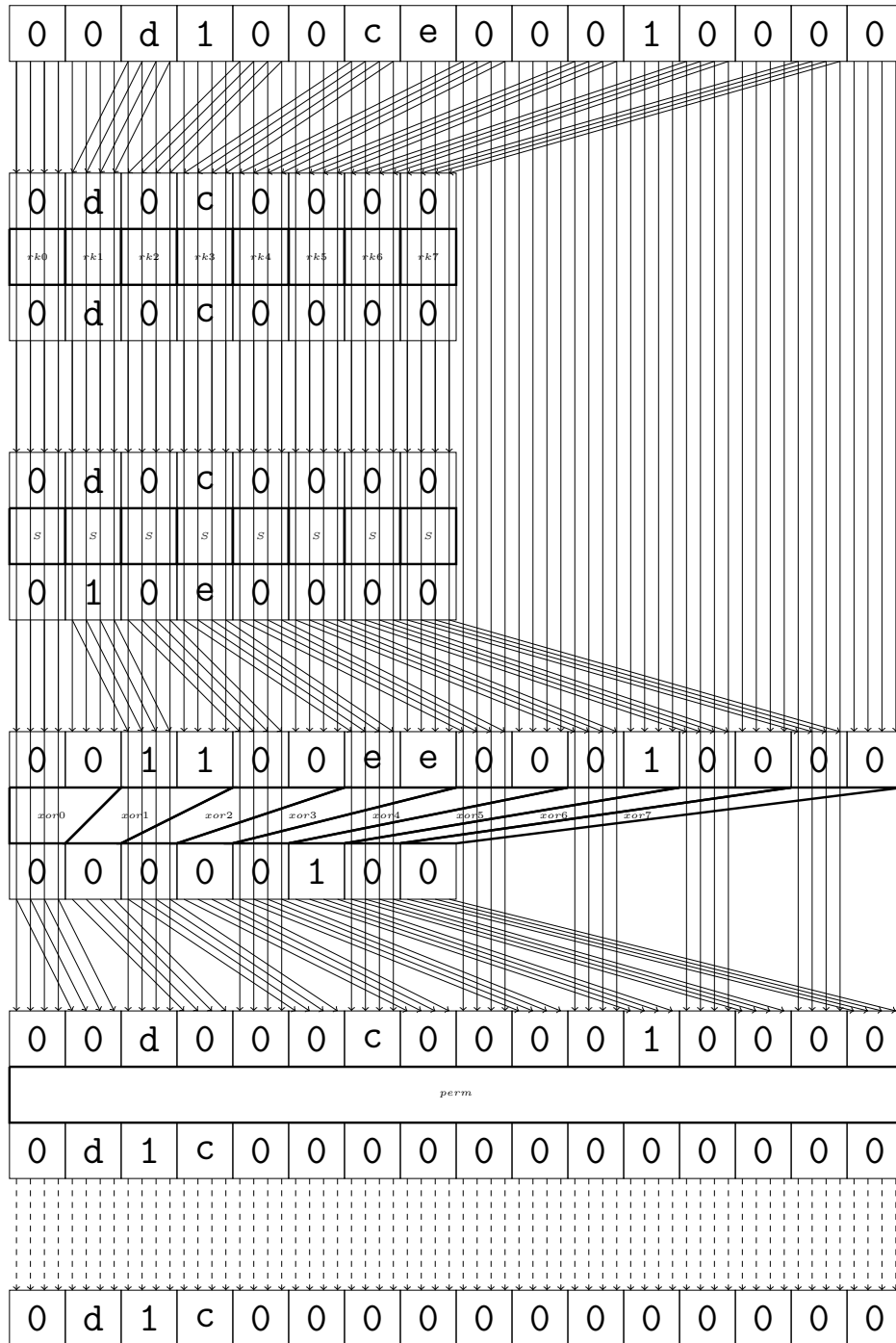
3 round

Maximal differential probability: $2^{-6.0}$



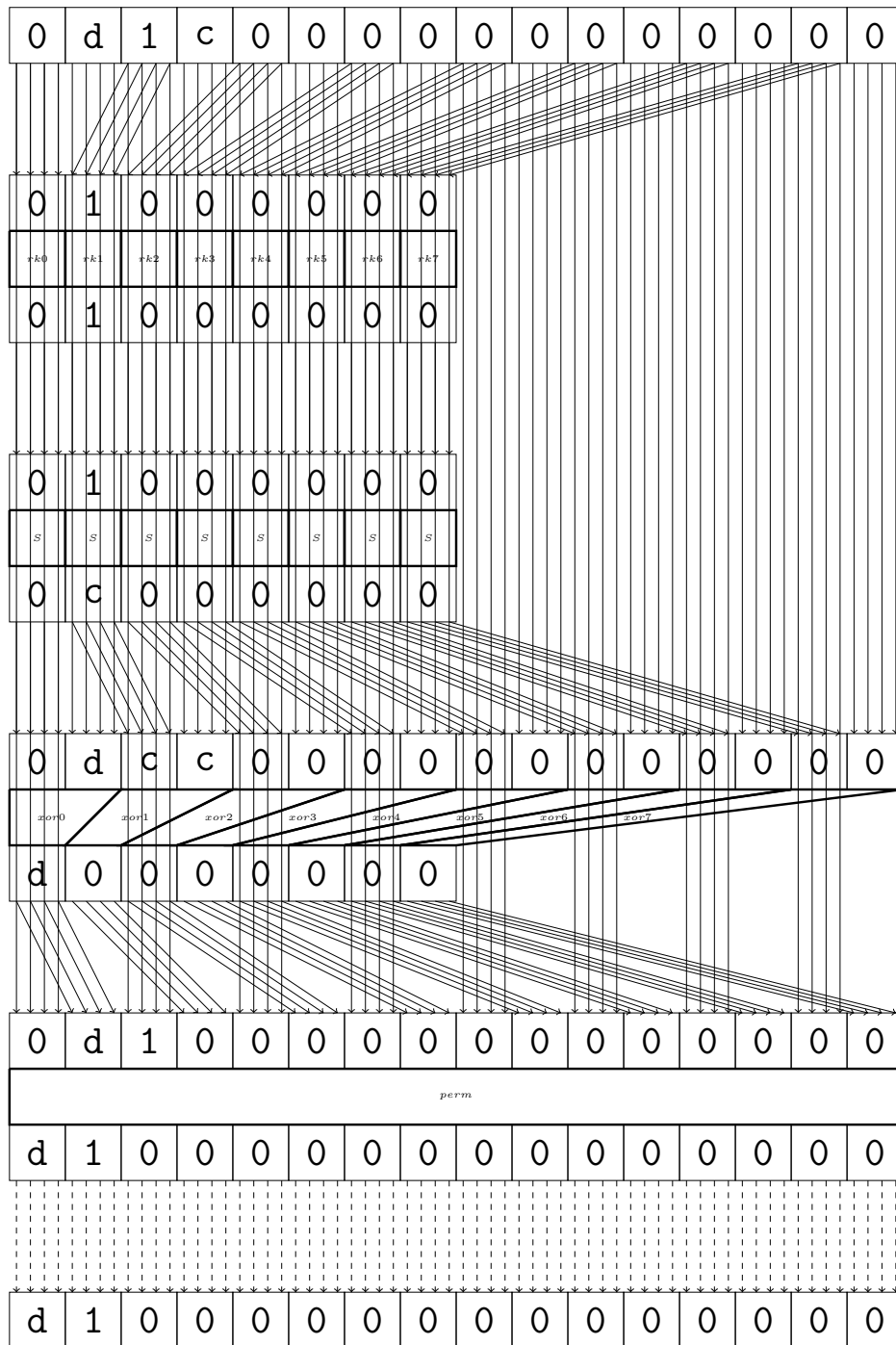
4 round

Maximal differential probability: $2^{-4.0}$



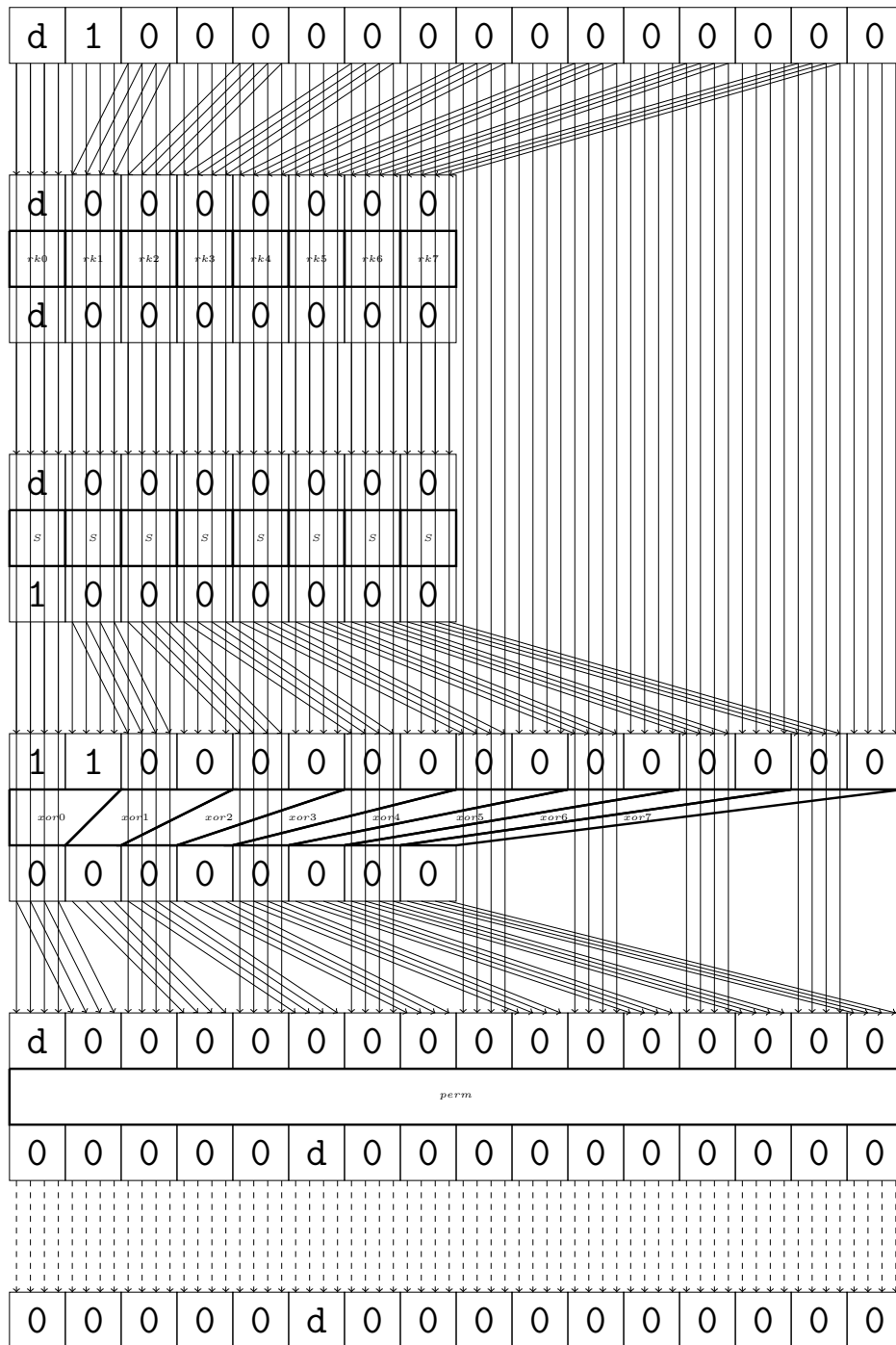
5 round

Maximal differential probability: $2^{-2.0}$



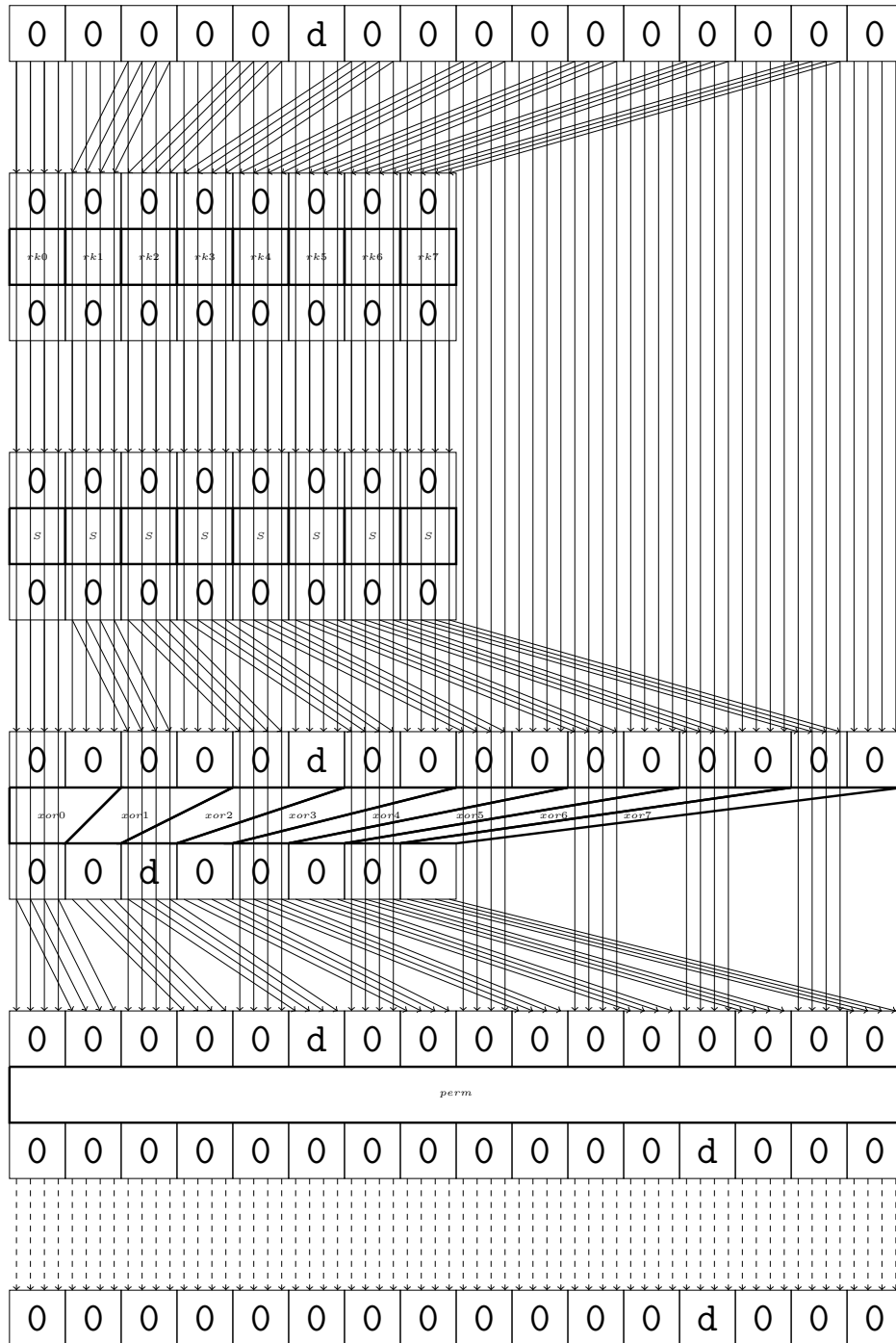
6 round

Maximal differential probability: $2^{-2.0}$



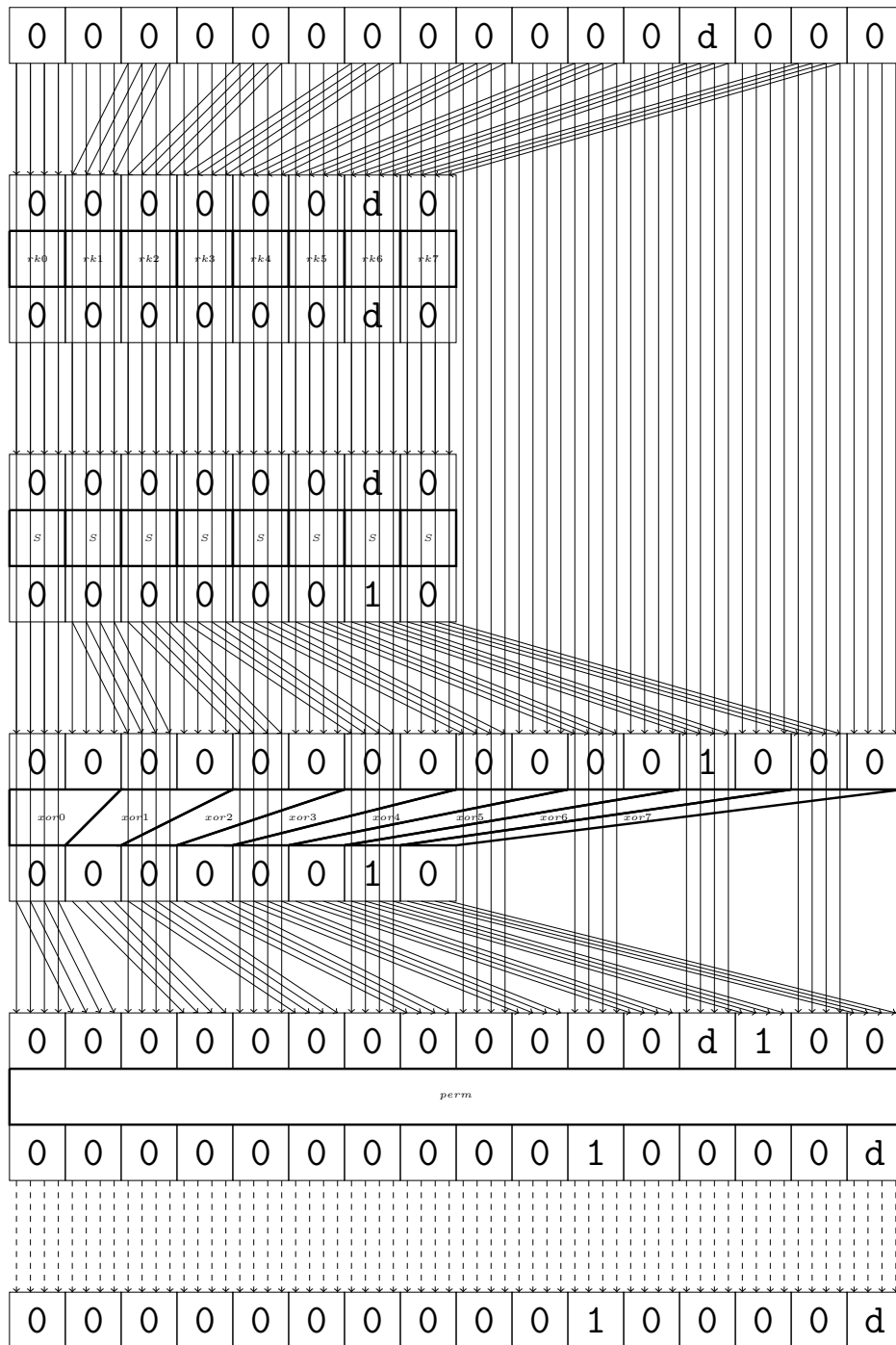
7 round

Maximal differential probability: $2^{-0.0}$



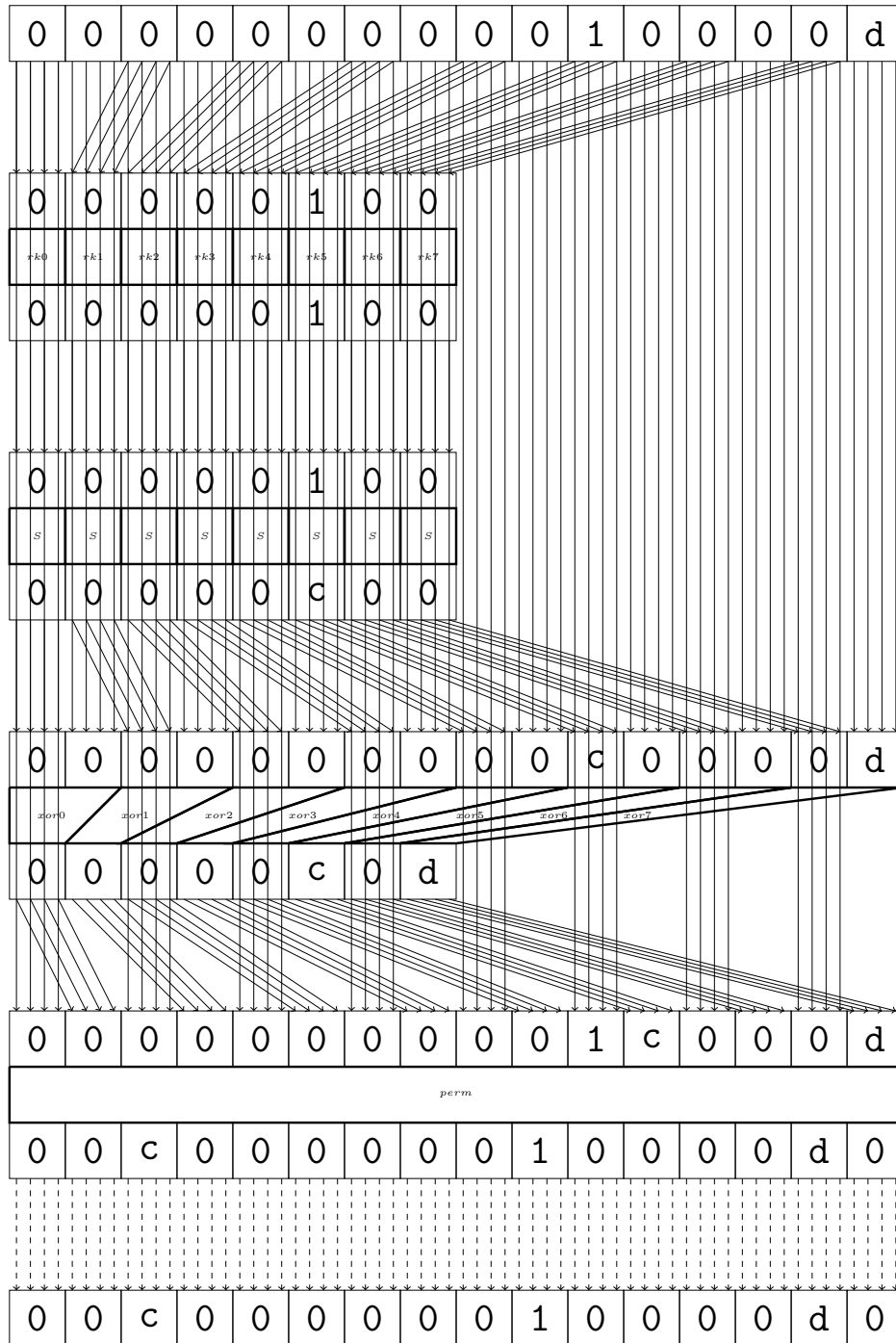
8 round

Maximal differential probability: $2^{-2.0}$



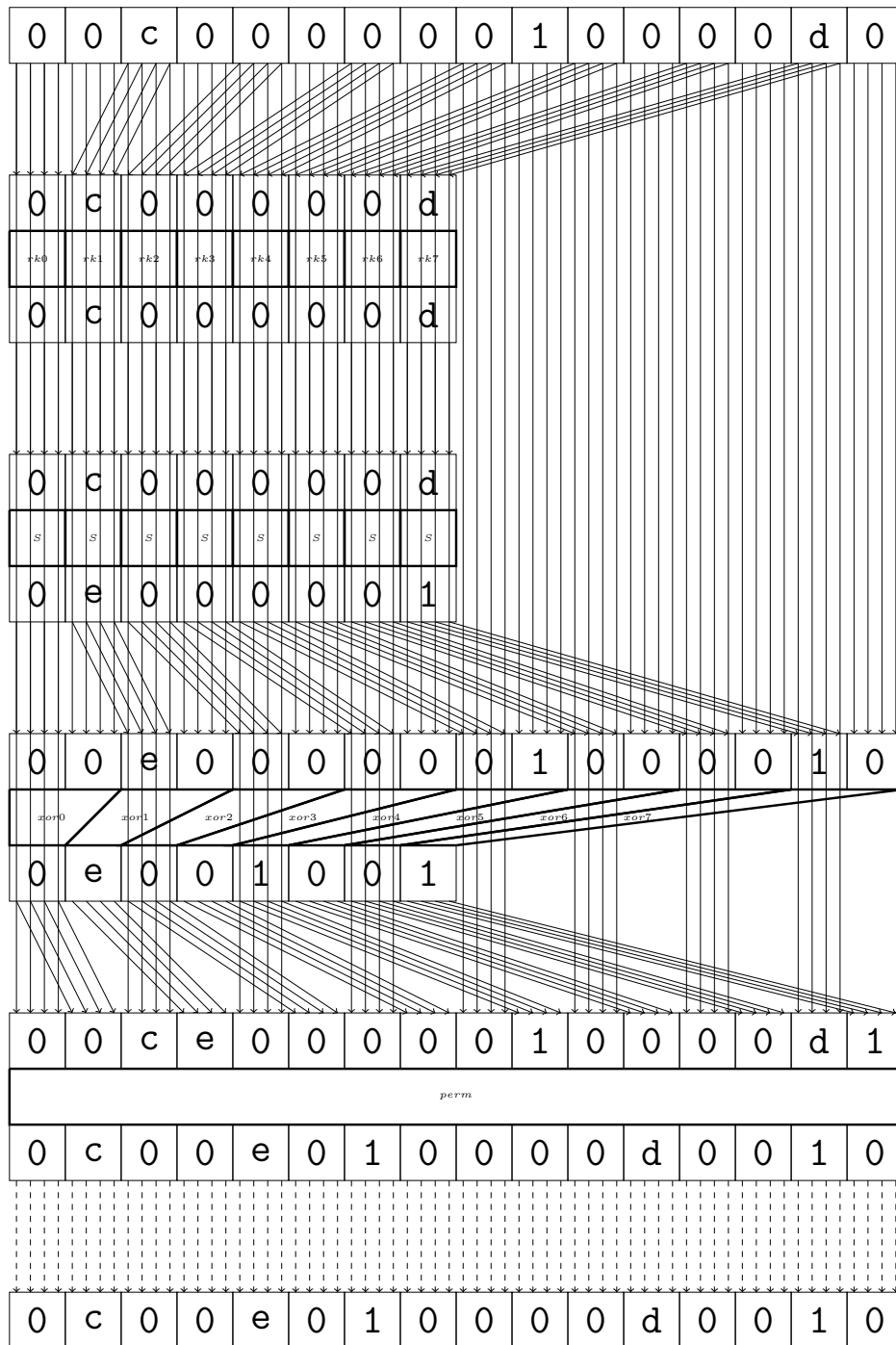
9 round

Maximal differential probability: $2^{-2.0}$



10 round

Maximal differential probability: $2^{-4.0}$



11 final

Maximal differential probability: $2^{-6.0}$

